Ovarian & Mullerian Infertility

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Block: Endocrine/Urology/ Reproduction

Thread: Clinical Medicine

Conflict of interest: None

Further Reading:

Taylor, Pal, & Seli. (2020). Clinical gynecologic endocrinology and infertility (9th ed.). Chapter 25: Female Infertility. Philadelphia: Wolters Kluwer Health/Lippincott Williams & Wilkins. (release of 10th ed 10/31/25)

Learning Objectives:

At the conclusion of this session, the participant should be able to:

1. Describe the impact of aging on (ovarian) fertility.
2. Define infertility.
3. List and describe the causes of (ovarian/mullerian) infertility.
4. Describe the evaluation of infertility.
5. Identify treatment options available.

Study Questions:

1. A 32-year-old cis-female nulligravida in a heterosexual relationship presents for evaluation. The couple has not used contraception for the last 15 months. She has regular menses occurring every 30-32 days. She has a history of chlamydia as a teenager which required hospitalization. Her partner is 39 and had not previously contributed to a pregnancy. Her examination is unremarkable.
    a. What is the most likely cause of this couple's infertility?
    b. What are the next steps in the evaluation of this couple?
    c. What treatment recommendations would you make?

2. A 27-year-old nulligravida cis-female in a same sex marriage presents for evaluation. She was placed on oral contraceptives (OCP) at the age of 15

for cycle control. She discontinued the OCP approximately 9 months ago. She has had 2 menses since that time. Her BMI is 23.9. The remainder of the examination is remarkable for mild hirsutism. The couple is interested in achieving a pregnancy.

a. What barriers exist to achieving pregnancy in this couple?
b. What are the next steps in the evaluation of this couple?
c. What treatment recommendations would you make?

3. A 44-year-old cis-female with 2 children ages 12 and 14 from a prior relationship is now remarried and presents for evaluation. They have not used contraception for the last 6 months; she is confused and disappointed as her prior pregnancies were conceived easily. She has regular menses occurring every 26-28 days. She has no history suggestive of pelvic surgery or infection. Her partner is 39 and has 2 children from a prior relationship. Her examination is unremarkable.

    a. What is the most likely cause of this couple's infertility?
    b. What are the next steps in the evaluation of this couple?
    c. What treatment recommendations would you make?

Aging and Fertility

1. Fertility declines with advancing ovarian age
    a. Demonstrated in studies of "natural" populations that do not restrict conception
    b. Supported by studies of conception rates among cis-females attempting pregnancy with donor sperm
    c. Peak fertility occurs between 20-24 years of age, with an initial slow and then more rapid decline
        i. Ages 25-29: 4–8% decrease
        ii. Ages 30-34: 15–19% decrease
        iii. Ages: 35-39: 26–46% decrease
        iv. 40 and over: as much as a 95% decrease
    d. Miscarriage rate increases with advancing age as well
2. Physiology of reproductive aging
    a. Follicular depletion
        i. At 16-20 weeks of pregnancy there are 6–7 million oogonia present
        ii. At birth there are 1-2 million oocytes (oogonia that have undergone 1st meiotic division)

1. Result of gene-regulated apoptosis iii. At the onset of puberty, there are 300,000 oocytes

iv. During reproductive life only about 400 oocytes are ovulated
v. The remainder are lost through atresia
vi. The rate of decline in oocyte number increases with increasing age

b. Oocyte aging
   i. Oocyte quality directly correlated with oocyte number (and hence age)
   ii. Rate of oocyte and embryo aneuploidy increases with increasing age
      1. Results from disordered regulatory mechanisms governing meiotic spindle formation and function
c. Endocrinology of reproductive aging
   i. Follicle stimulating hormone (FSH) levels rise, especially in the early follicular phase
      1. Results from a progressive decrease in negative feedback from the smaller cohort of follicles

Fertility/Infertility

1. Maximizing fertility (assuming everything works!)
   a. Lifestyle modification
      i. Establishing a normal weight
      ii. Smoking cessation
      iii. Avoiding excessive alcohol consumption
      iv. Avoiding excessive caffeine
      v. Avoiding substance abuse
   b. Coital timing
      i. Usually 2-3 times/week is adequate (assuming spacing)
         1. Sperm in the reproductive tract can survive and likely fertilize an oocyte for 3-5 days
         2. An oocyte can be fertilized for 12-24 hours after ovulation
2. Definition of infertility
   a. Inability to achieve a pregnancy within one year of attempting conception in cis-females who are older than 35 and/or those who have previously been pregnant (secondary infertility)
      i. The definition was established because approximately 80% will be successful within 1 year of attempting pregnancy

Causes of Infertility

1. Incidence varies based on the population studied (numbers below will not add up to 100% and are ranges)
    a. Age
    b. Duration of infertility
    c. Level of care-specialist vs generalist
    d. Many couples will have more than 1 cause
2. Testicular factor
    a. Low sperm count, motility and/or morphology
    b. 20-40%
    c. Increases with age
3. Ovulatory
    a. Irregular or no periods
    b. 20-40%
    c. More common in younger individuals
4. Tubal and peritoneal factors
    a. Endometriosis, tubal scarring/blockage secondary to endometriosis, surgery or infection
        i. Can be proximal or distal
    b. 30-40%
    c. Not significantly impacted by age
5. Uterine factor
    a. Uterine fibroids, scarring, polyps, congenital abnormalities
    b. Less than 10%
    c. Not significantly impacted by age
6. Cervical factor
    a. Scarring, abnormalities of cervical mucus
    b. Less than 5%
    c. Not significantly impacted by age
7. Unexplained infertility
    a. Should this include age related infertility??
    b. 15-25%
    c. Increases with age

Evaluation of Infertility

1. Intended to evaluate the etiology of the infertility
    a. All potential causes must be evaluated

2. History
    a. Ob/gyn history
        i. Menstrual history
        ii. History of sexually transmitted infections
        iii. History of abnormal pap smears
        iv. Obstetrical history
        v. Coital frequency
    b. Duration of infertility
        i. Prior treatment
    c. Medical history
        i. Medications ii. Allergies
        iii. Surgical history
    d. Social history
        i. Tobacco
        ii. Alcohol
        iii. Caffeine iv. Recreational drugs
        v. Exercise
        vi. Occupational exposures
    e. Family history
        i. Birth defects, mental retardation
        ii. Early menopause or reproductive failure
3. Physical exam
    a. BMI
    b. Thyroid
    c. Breast discharge
    d. Signs of androgen excess
    e. Abnormalities on pelvic exam
        i. Fibroids
        ii. Ovarian enlargement
        iii. Tenderness
4. Testing
    a. Sperm
        i. Semen analysis
    b. Ovarian function
        i. Ovarian reserve
            1. Anti-mullerian Hormone (AMH)
            a. Produced by granulosa cells of preantral and small antral follicles

b. Correlates with response to ovarian stimulation, oocyte yield at IVF, embryo quality and pregnancy rate
c. Less valuable in the general population

2. Antral follicle count
a. Total number of follicles between 2-10 mm
b. Proportional to the number of primordial follicles remaining in the ovary
c. Correlates with response to ovarian stimulation

3. FSH
a. Varies across the cycle so should be obtained d 2-5 of the menstrual cycle
b. Varies from cycle to cycle with highest value being most predictive of outcome
c. Inversely correlated with ovarian reserve
d. Estradiol, which suppresses FSH production via negative feedback, should be obtained simultaneously

c. Ovulation
i. Regular menses presumptive evidence of ovulation
1. Can be confirmed by mid-luteal progesterone level ii. An/oligo-ovulation
1. Characterized by menstrual interval >35 days
2. Etiology
i Polycystic ovarian syndrome (PCOS)
ii Thyroid dysfunction
iii Hyperprolactinemia
iv Adrenal hyperfunction
v Hypothalamic amenorrhea
vi Other
3. Evaluation
a. TSH, prolactin
b. Androgens if evidence of hyperandrogenism
i DHEAS, testosterone, 17-hydroxyprogesterone
ii Ultrasound

d. Utero-tubal anatomy
i. Ultrasound
ii. Hysterosalpingography or sonohysteroasalpingography
iii. Laparoscopy
1. Best method to diagnose endometriosis

a. Present in 20-30% percent of infertile individuals (data varies based on population studied)
b. Minimal impact on pregnancy rate

iv. Hysteroscopy

1. Gold standard for evaluating uterine cavity
   a. Polyps
   b. Fibroids
   c. Adhesions
   d. Congenital abnormalities

v. Cervical mucus abnormalities

1. Rare cause of infertility

a Evaluation not done as a routine; Assessment of cervical mucus has poor positive predictive value

Treatment Options

1. Ovulatory Dysfunction
   a. Treat the underlying condition
   b. Ovulation induction
      i. Can be used for treatment of an or oligoovulation or empirically in individuals with normal ovulatory function
      ii. Oral medications 1. Clomiphene citrate

   a. Selective estrogen receptor modulator (SERM)
   b. May be better for unexplained infertility
   c. Risk multiple pregnancy <10%
      i. Rarely more than twins
      d. Side effects
      i. Hot flashes, headaches, mood changes, visual disturbance
2. Letrozole
   a. Aromatase inhibitor
   b. May be better for ovulatory dysfunction
   c. Risk of multiple pregnancy <10%
      i. Rarely more than twins

d. Side effects
i. Hot flashes, headaches, mood changes, visual disturbance
ii. Typically tolerated better than clomiphene

3. Metformin
a. Oral anti-hyperglycemic agent
b. Rarely used to treat ovulatory dysfunction

iii. Gonadotropin therapy
1. FSH (or FSH/LH) injections
2. Risks
a. Multiple pregnancy 15-20%
i. 2-3% high order multiples
b. Ovarian hyperstimulation syndrome
i. Characterized by significant ovarian enlargement, ascites, hemoconcentration
3. Side effects
a. Tolerated well

iv. Ovarian drilling (laparoscopy)
1. Rarely used
a. Invasive
b. Risk of causing adhesions

2. Tubal disease
a. Surgery
i. Laparoscopy
1. Tubal repair
a. Typically not effective
b. Tubal Canalization
i. Patency established >50%, pregnancy rate <50% of those

c. IVF
i. Treatment of choice

3. Endometriosis
a. Medication
i. GnRH analog (Lupron)
ii. Progesterone
iii. Oral contraceptive
iv. Orlissa

b. Surgery
i. Does not appear to be effective in improving pregnancy rate

4. Uterine Abnormalities

a. Hysteroscopy
    i. Myomectomy
    ii. Polypectomy
    iii. Lysis of adhesions
b. Laparoscopy/laparotomy
    i. Myomectomy
c. Repair congenital abnormality

5. Unexplained infertility
    a. Ovarian stimulation
        i. Clomiphene citrate
        ii. Gonadotropin therapy iii. IVF

Answers to Study Questions:

These answers are not meant to be all encompassing; rather they are examples of the types of thought processes needed for application of the material.

1. The couple has been attempting pregnancy for over a year and would therefore be considered infertile. She is young and likely has good ovarian reserve. Her cycles are quite regular and she is likely ovulatory. She has a history of chlamydia and as a result is at greater risk for tubal disease.

   Though you suspect tubal disease, the evaluation should be complete and should include a semen analysis, assessment of ovarian reserve, mid-luteal progesterone and hysterosalpingogram (HSG). Given the history of chlamydia she should be given prophylactic antibiotics for the HSG.

   As always treatment is dependent on testing. Assuming normal semen analysis and HSG, and adequate ovarian reserve, the recommendation

   would be for empiric ovulation induction and intrauterine insemination (IUI). IVF would be recommended if tubal disease was present.

2. This patient has irregular cycles and is likely anovulatory. In addition, she will require donor sperm to achieve a pregnancy.

   The first step would be to evaluate the cause of the menstrual irregularity. In addition, HSG could be recommended (to rule out any abnormalities).

   In the absence of abnormal testing, oral ovulation induction would be required. Insemination with donor sperm would be performed.

3. The couple has been attempting pregnancy for 6 months and would therefore be considered infertile. Even though she has regular cycles and is likely ovulatory, the couples lack of conception is likely an age-related oocyte factor.

   Despite your suspicions, the evaluation should be completed and should include a semen analysis, assessment of ovarian reserve and hysterosalpingogram (HSG).

   As always treatment is dependent on testing. Assuming normal semen analysis and HSG, and adequate ovarian reserve numbers, the options would include empiric ovulation induction and intrauterine insemination (IUI), in vitro fertilization (IVF), oocyte donation and not engaging in care. There are pros and cons (risks/benefits) to each approach and require discussion and shared decision making.